Global Environmental Picture

Major Global Environmental Concerns

Population Growth

Soil Degradation

Atmospheric Change

Biodiversity Loss

Global Environmental Concerns

Population Growth





Biodiversity Loss

Atmospheric Change





Soil Degradation

Global Environmental Concerns

Population Growth

Population

Total human inhabitants of a specified area, such as a city, country, or continent, at a given time. The study as a discipline is known as demography.

Deals with

It is concerned with the size, composition, and distribution of populations; their patterns of change over time through births, deaths, and migration; and the determinants and consequences of such changes.

Why do you need to know?

- Knowledge of population studies is important for planning, particularly by governments, in fields such as health, education, housing, social security, employment, and environmental preservation.
- Such studies also provide information needed to formulate government population policies, which seek to modify demographic trends to achieve economic and social objectives.

Global Environmental Concerns

Population Growth

Robert McNamara(1973), wrote *One Hundred Countries, Two Billion People*

Rapid population growth was a threat that would have "catastrophic consequences"

James Wolfensohn,
World Bank president from 1995 to 2005

L

Robert Zoellick delivered a speech identifying six strategic themes to meet global challenges. Reducing population was not one of them; in fact, "population growth" is not mentioned even once

Robert McNamara, *One Hundred Countries, Two Billion People* (New York: Praeger, 1973), p. 31.

Global Environmental Concerns

Population Growth

Last Ice Age, about 13,000b years ago, when humans on all continents were still living as hunter–gatherers, or 12,000b years ago with the first signs of agricultural settlements, or 7,000m years ago with the first indications of urbanization.

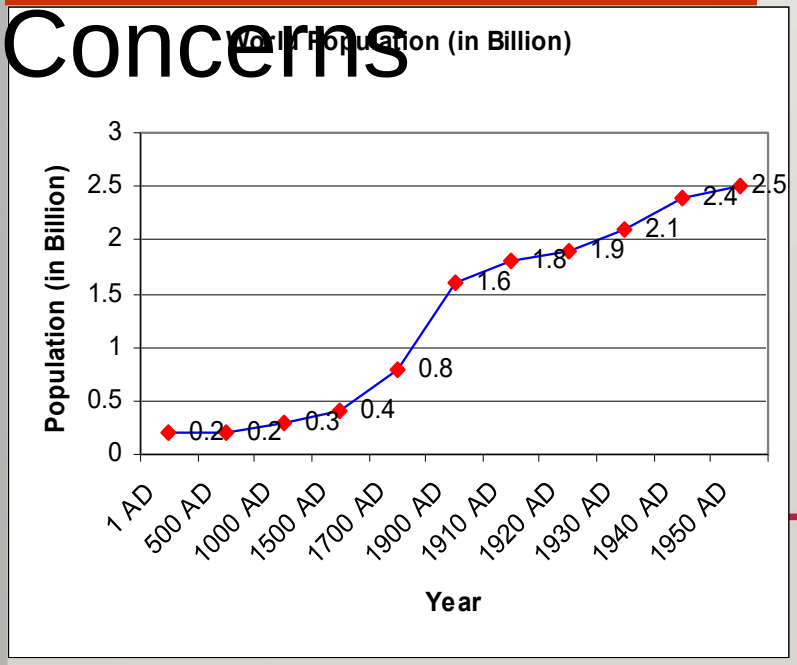
From the beginning of human settlements, it took more than 10,000 years for the world's population to reach 1 billion in 1820 1 billion people every 12 to 15 years. No wonder McNamara was so worried.

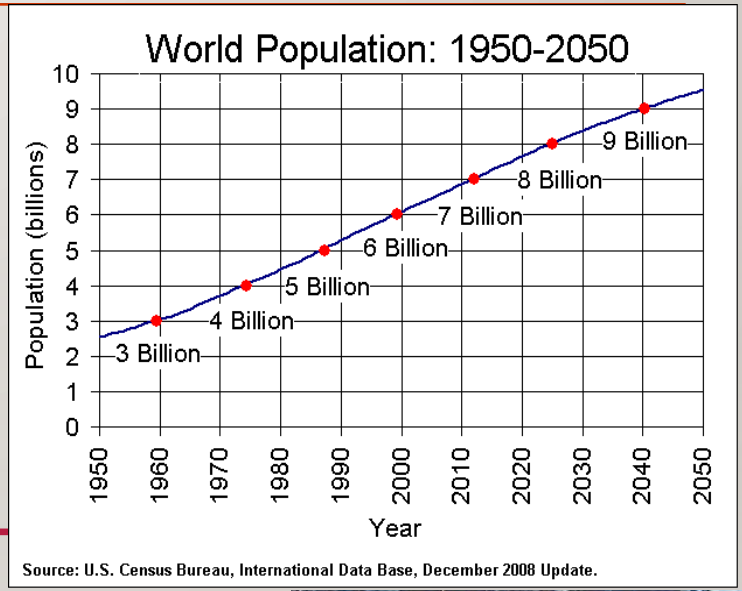
During the Industrial Revolution (late eighteenth and early nineteenth centuries), the food supply grew and became more reliable. The death rate fell, life expectancy increased, and population growth gradually accelerated.

The Industrial Revolution, which marked the start of modern economic growth, further expanded the Earth's **population-carrying capacity**.

Global Environmental Concerns

Populatio
n Growth



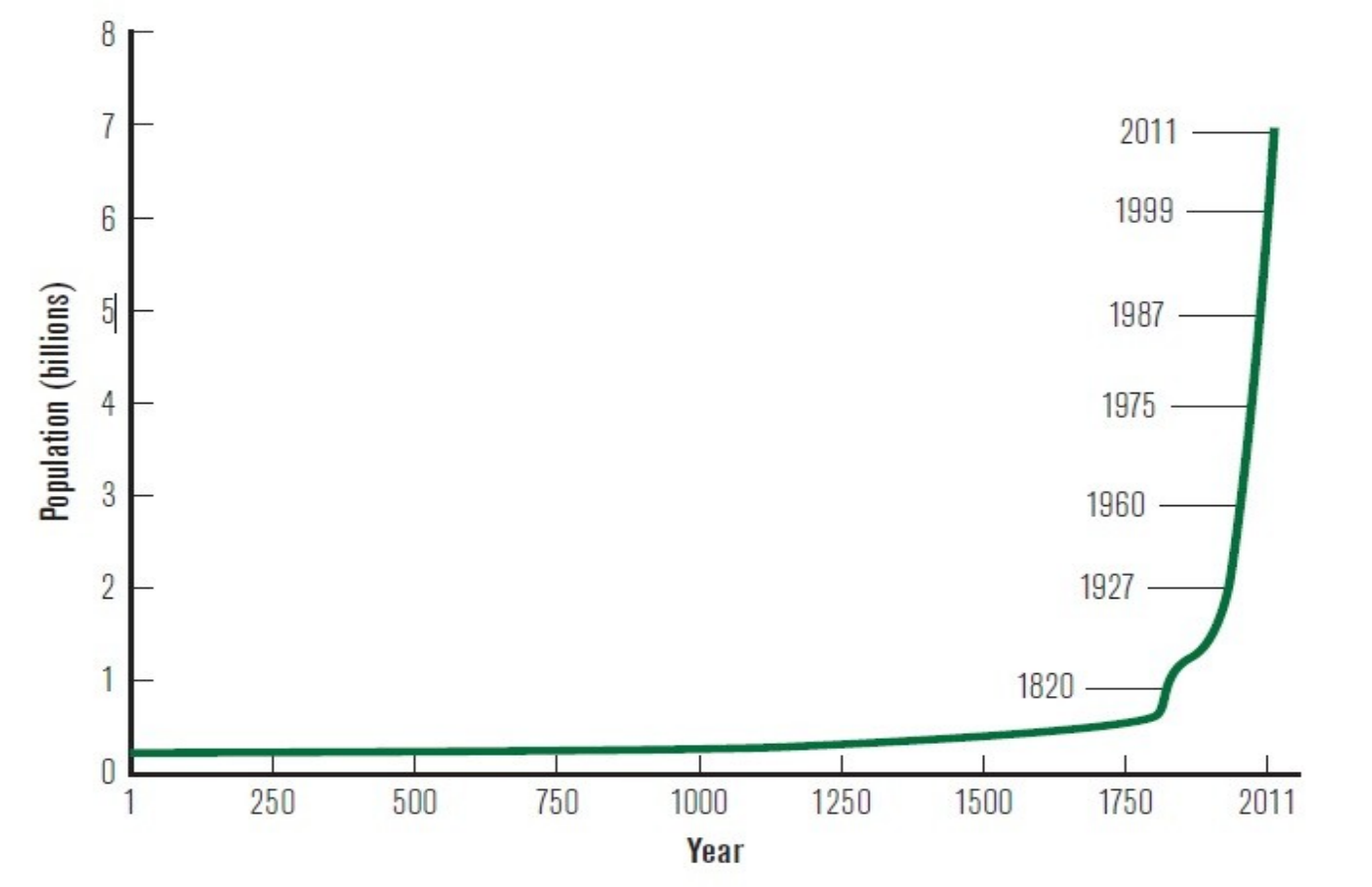




- four or five children are born in every second
- two other people die in same time
- nearly 2.5 more humans per second added to the world population.
- This means, we are growing almost 83 million more people per year.

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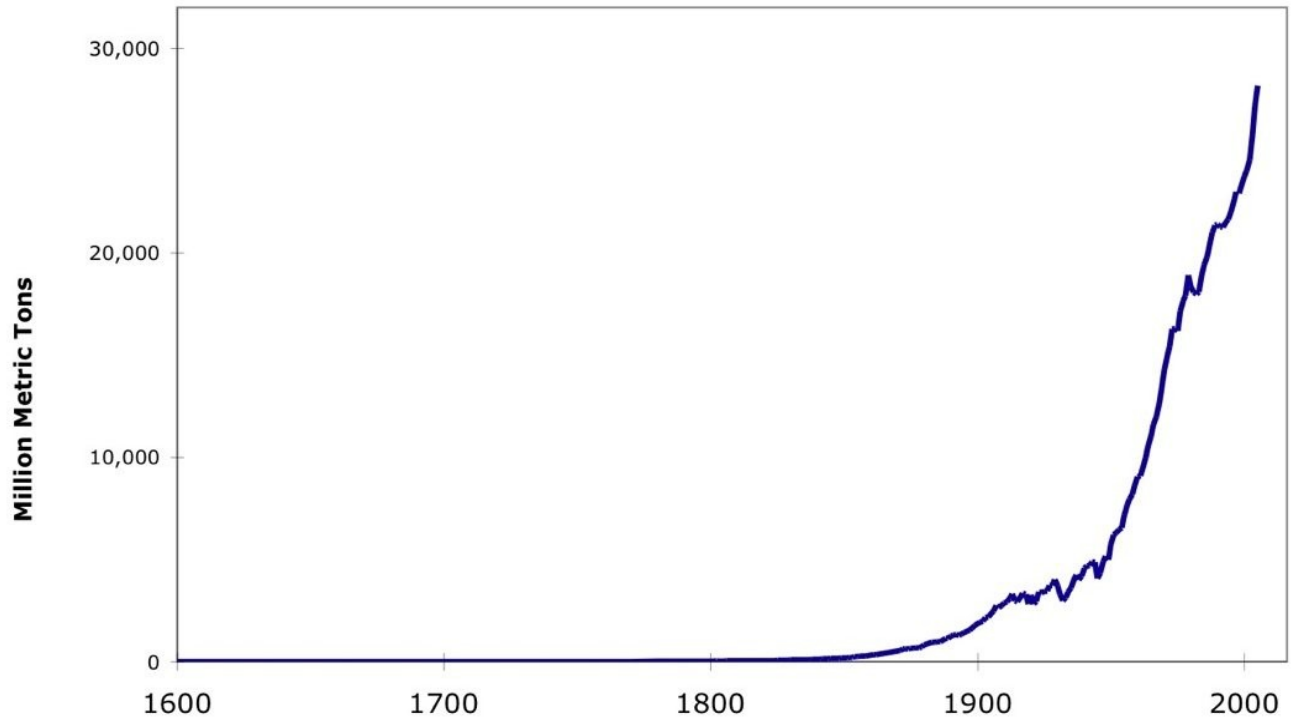
Populatio
n Growth



Don’t know what made this chart explode since 1900. Charts show exponential growth and are one of the major causes of Global Warming.

Global Environmental Concerns

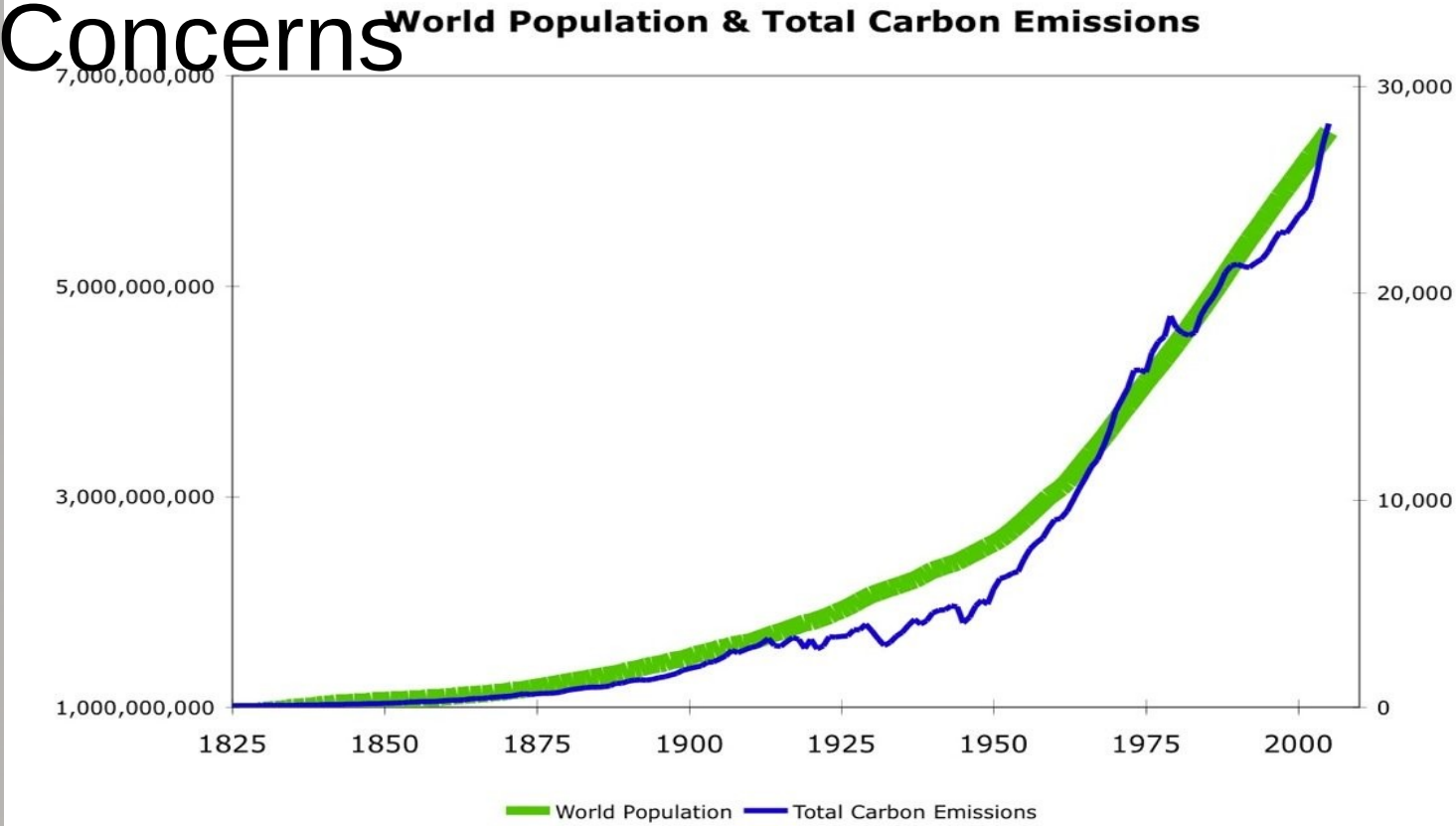
Populatio
n Growth



A significant Hockey Stick. Worldwide Total Carbon Emissions have experienced exponential growth. The chart shows historic Total Annual Worldwide Carbon Emissions since 1600.

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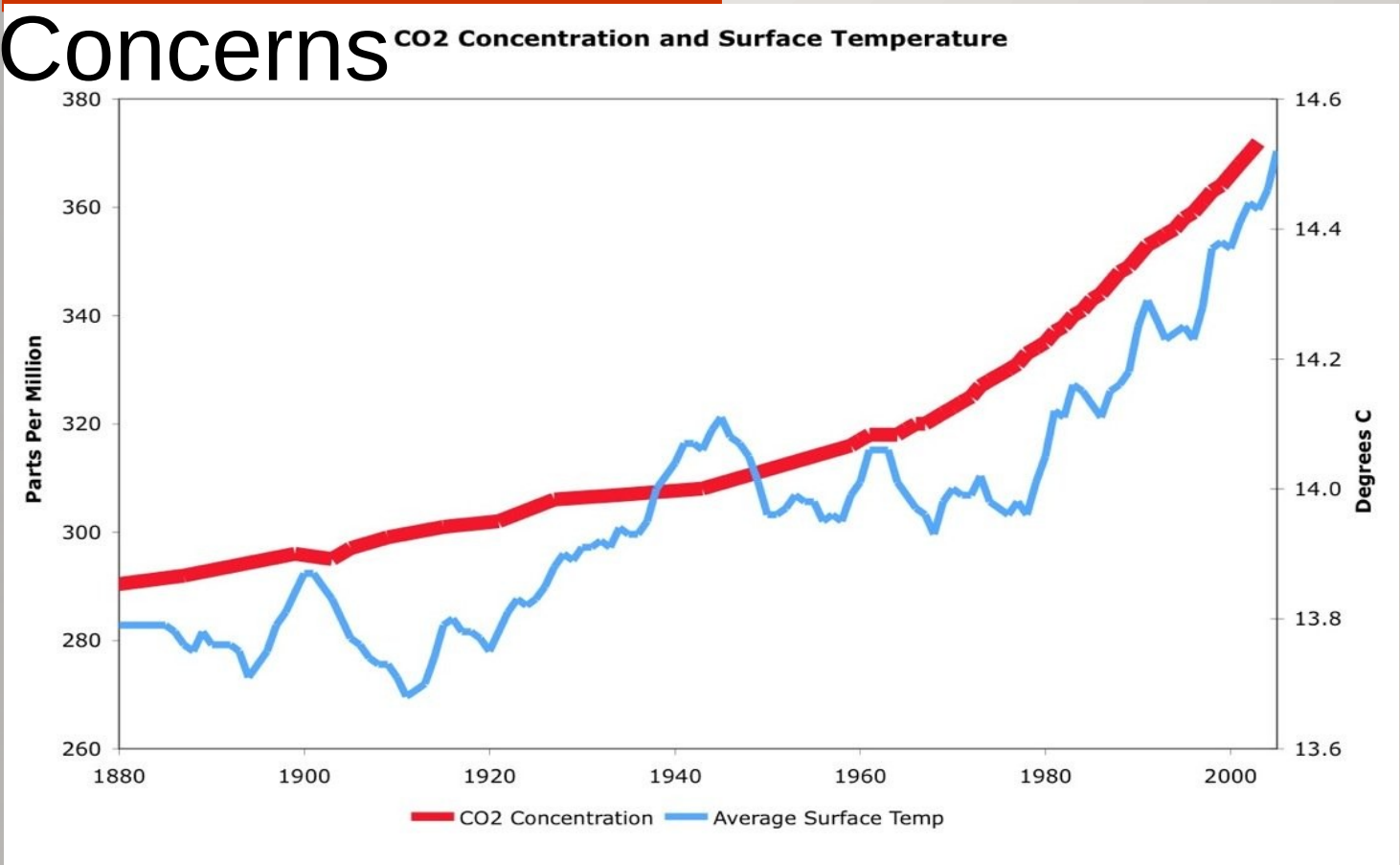
Populatio
n Growth



As Al Gore said, “Yes, they fit.” This chart shows how Total Annual Carbon Emissions are directly affected by Population Growth.

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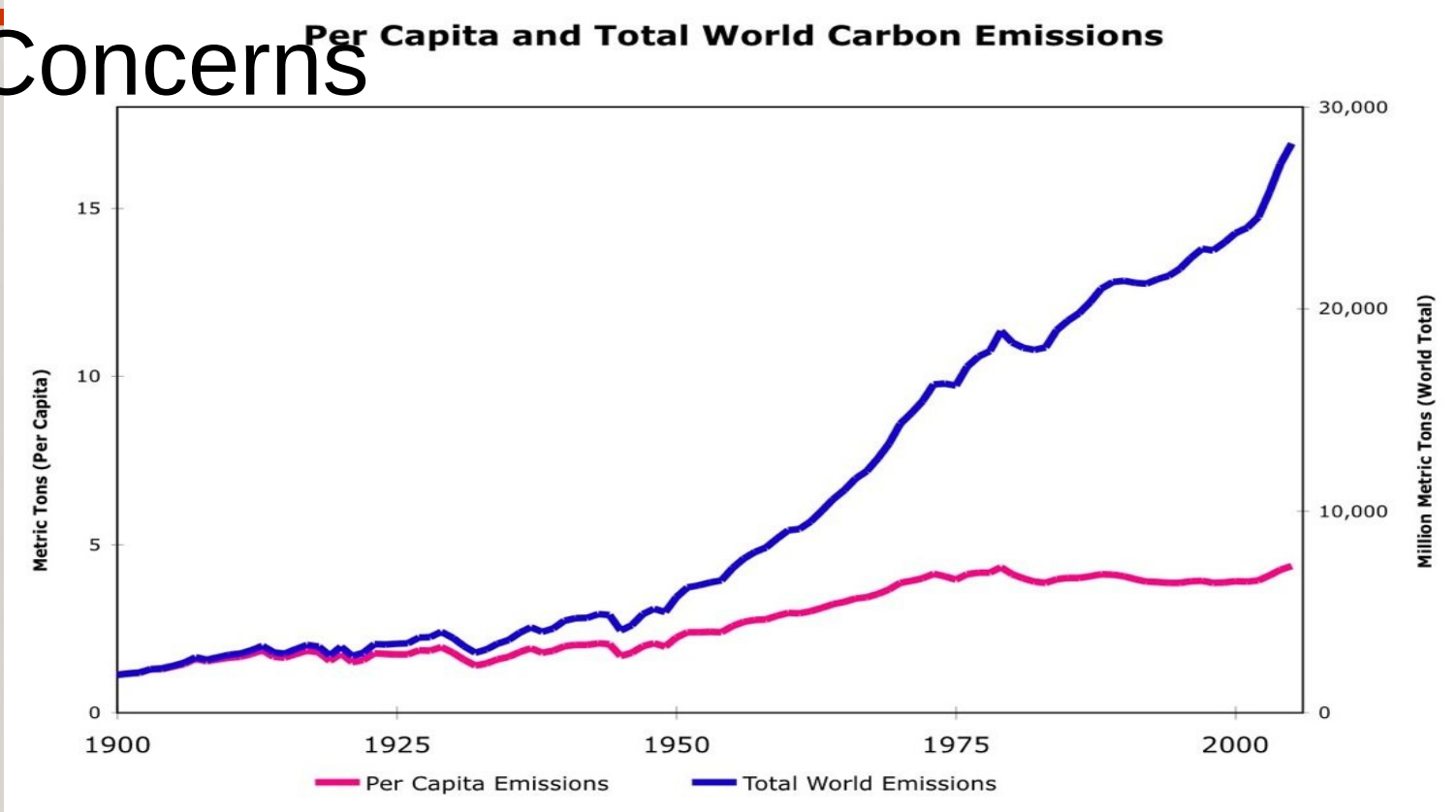
Populatio
n Growth



This chart shows the correlation between Atmospheric Carbon Concentration (in Parts Per Million) and Average Surface Temperature.

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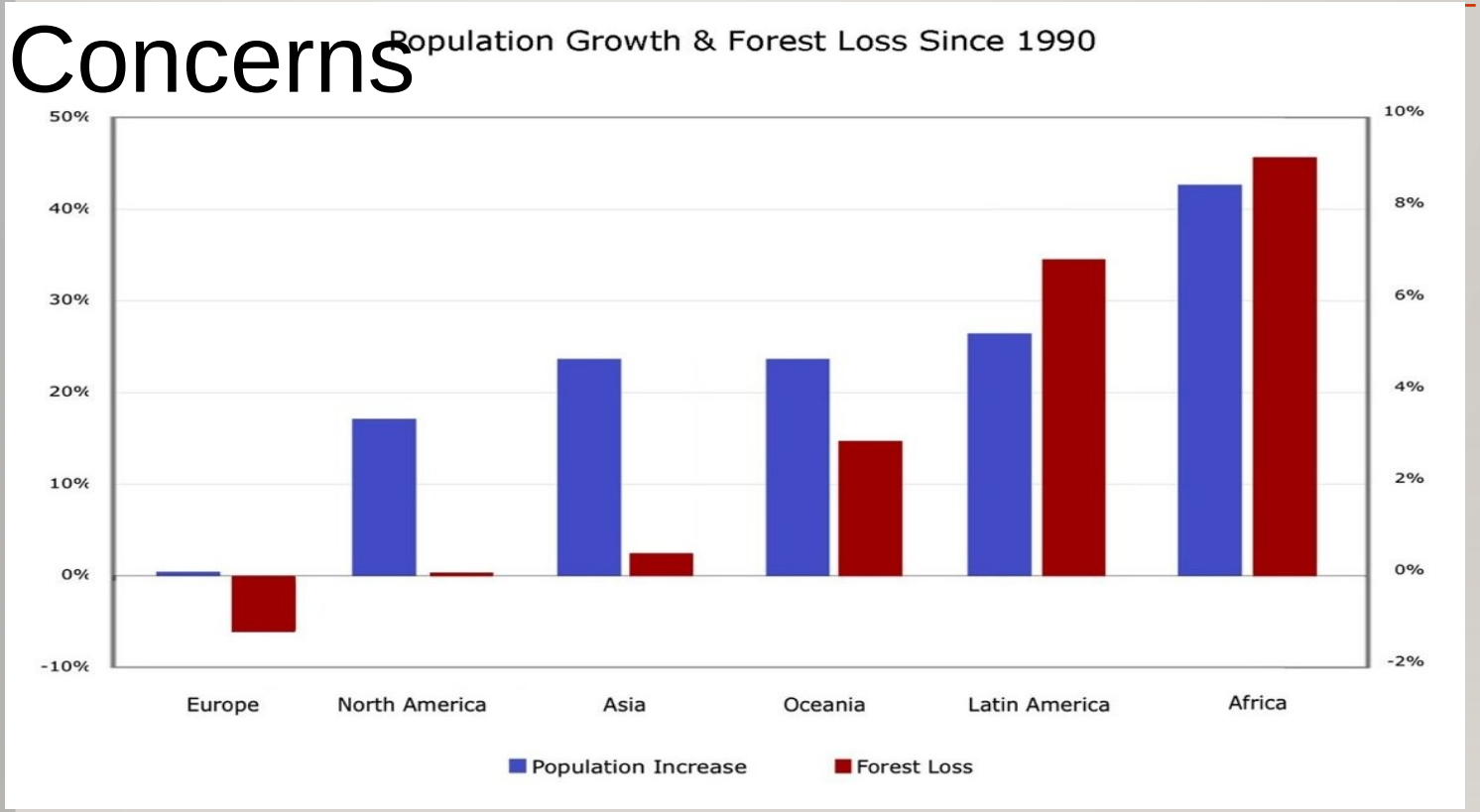
Populatio
n Growth



A curious case of divergence. This chart shows how Total Worldwide Carbon Emissions have increased at an Exponential Rate, while Per Capita Carbon Emissions have grown at a Linear Rate during the past century. To what can we attribute the difference in the rate of growth?

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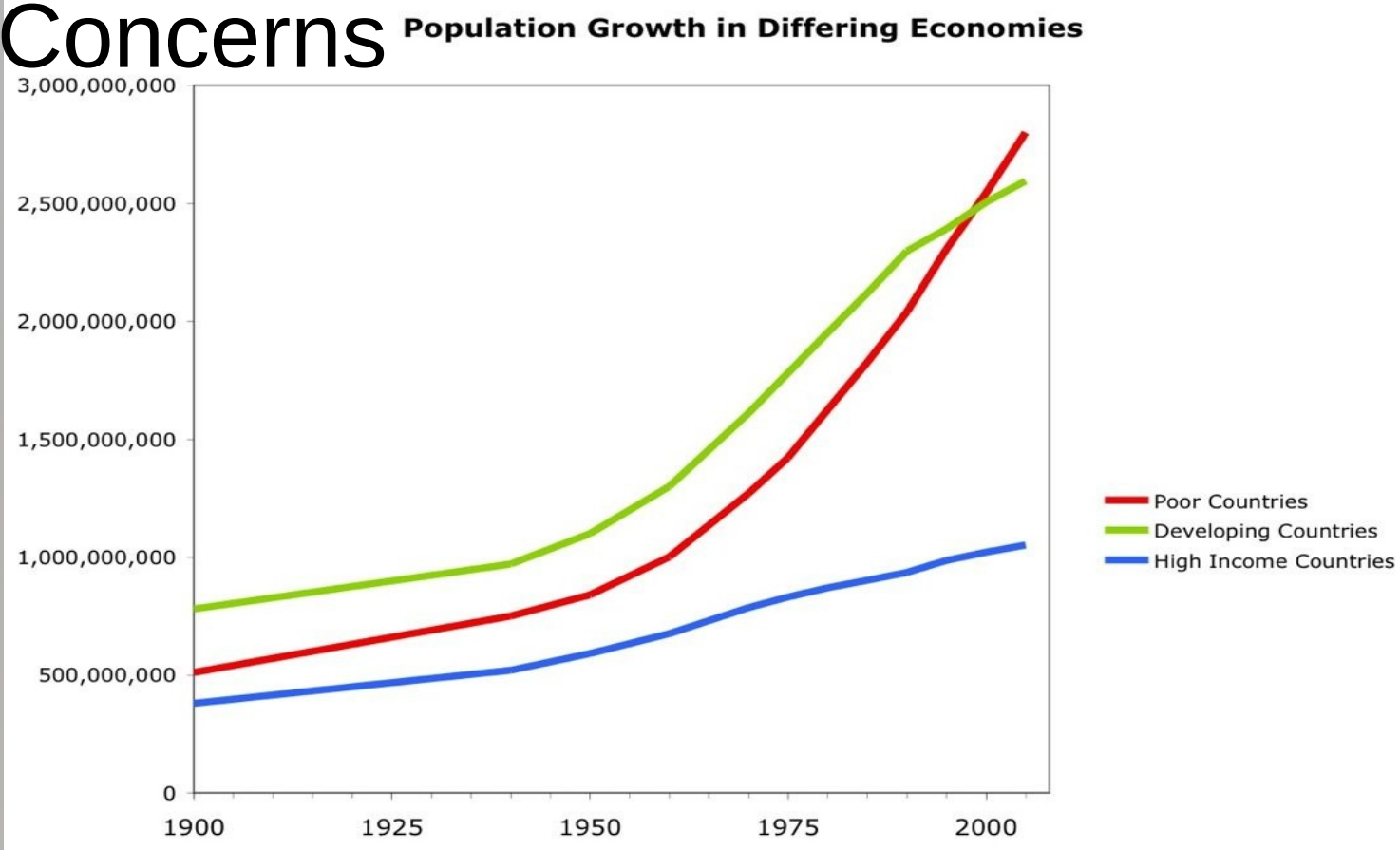
Populatio
n Growth



This very important chart shows the relationship between Population Growth and deforestation. Conclusion: The faster the population grows, the faster the forests disappear.

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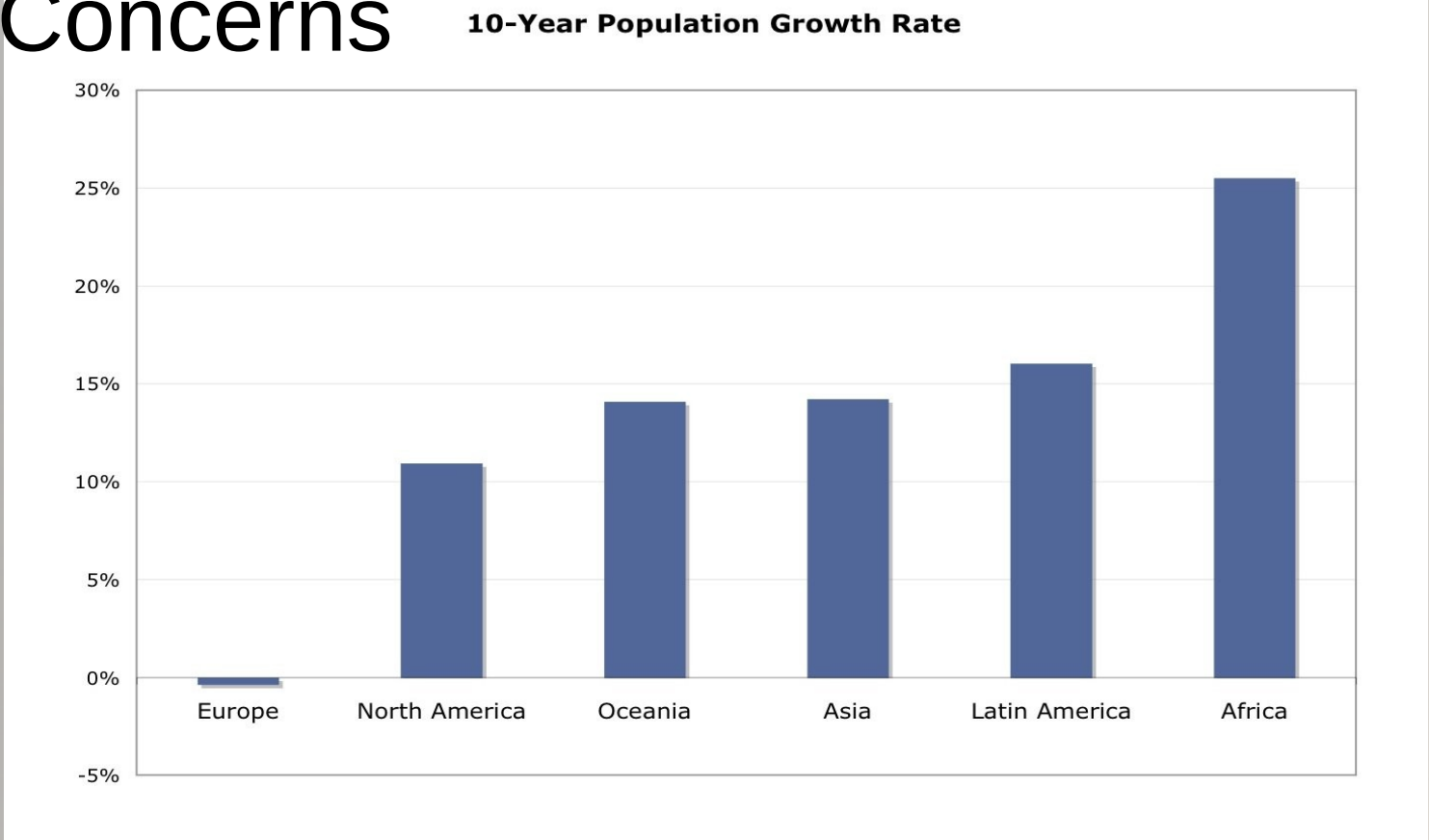
Populatio n Growth



This chart shows how Population Growth Rates are affected by Income.

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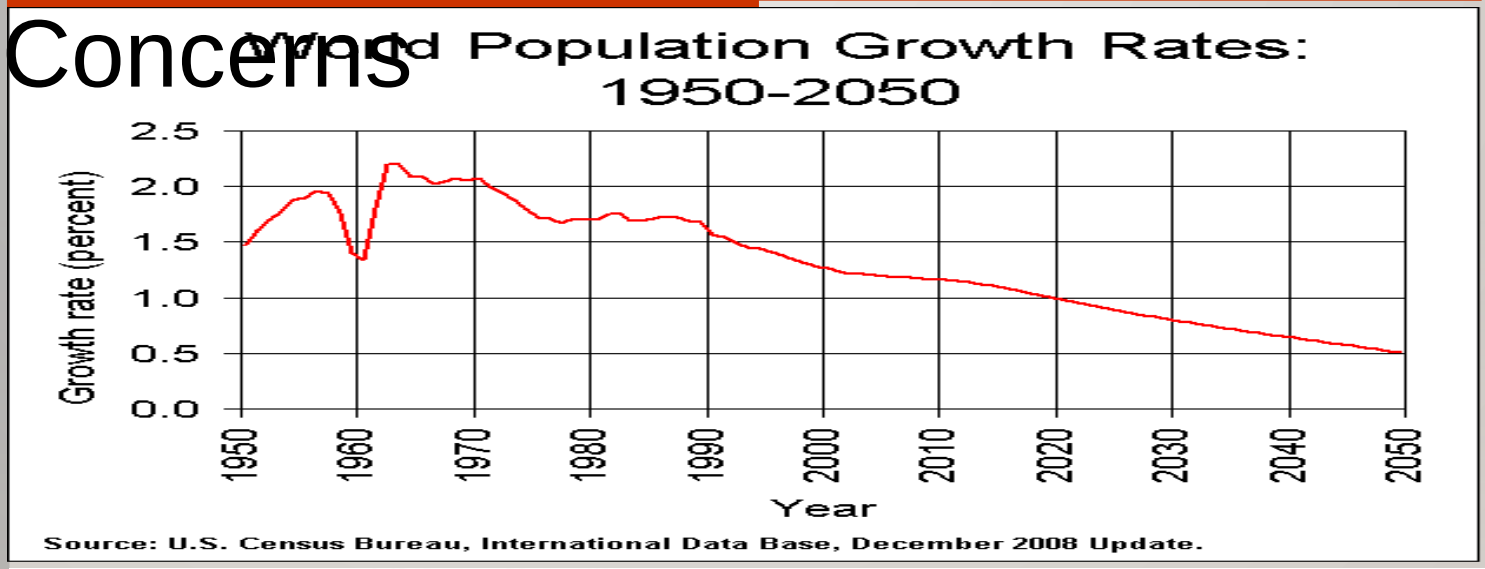
Populatio n Growth

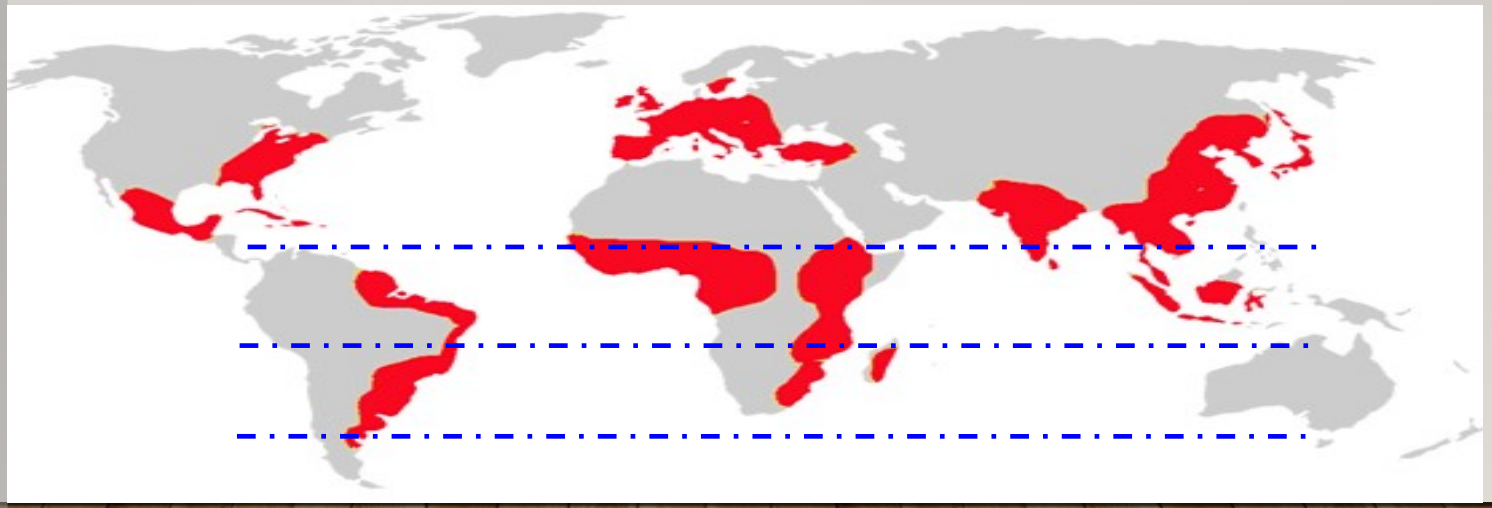


Asia has grown the most in absolute terms, and **Africa** leads the world in its **Rate of Growth**.

Global Environmental Concerns

Populatio n Growth





Global Environmental

Populatio
n Growth



Global Environmental Concerns

Populatio
n Growth

Impacts of Population Growth

- Great pressure is being placed on arable land, water, energy, and biological resources to provide an adequate supply of food while maintaining the integrity of our ecosystem.
- Overload the Earth's life-support systems
- Vital resources are stressed by the **dual demands** of increasing population and increasing consumption per person
-

- Groundwater supplies are being depleted, agricultural soils are being degraded, oceans are being overfished, oil reserves are being drawn down, and forests are being cut faster than they can regrow
- Human population inevitably outstrips their food supply and eventually collapses into starvation, crime, and misery
- Sometimes it is viewed that the root cause of environmental degradation is **inequitable distribution of wealth and power rather than population size**

Global Environmental Conc



Soil Degradation

… is a marvelous substance, a living resource of astonishing beauty, complexity, and frailty.

It is a complex mixture of weathered mineral materials from rocks, partially decomposed organic molecules, and a host of living organisms.

- ecosystem by itself



- essential component of the biosphere

Global Environmental Concerns

Soil Degradation

Soil Degradation

Based on both biological productivity and our expectations about what the land should be like. Often this is a subjective judgment, and it is difficult to distinguish between human-caused deterioration and natural conditions, like drought. We generally consider the land degraded when the soil is impoverished or eroded, water runs off or is contaminated more then normal, vegetation is diminished, biomass production is decreased, or wildlife diversity diminishes. It entails a decrease in soil's productivity and quality.

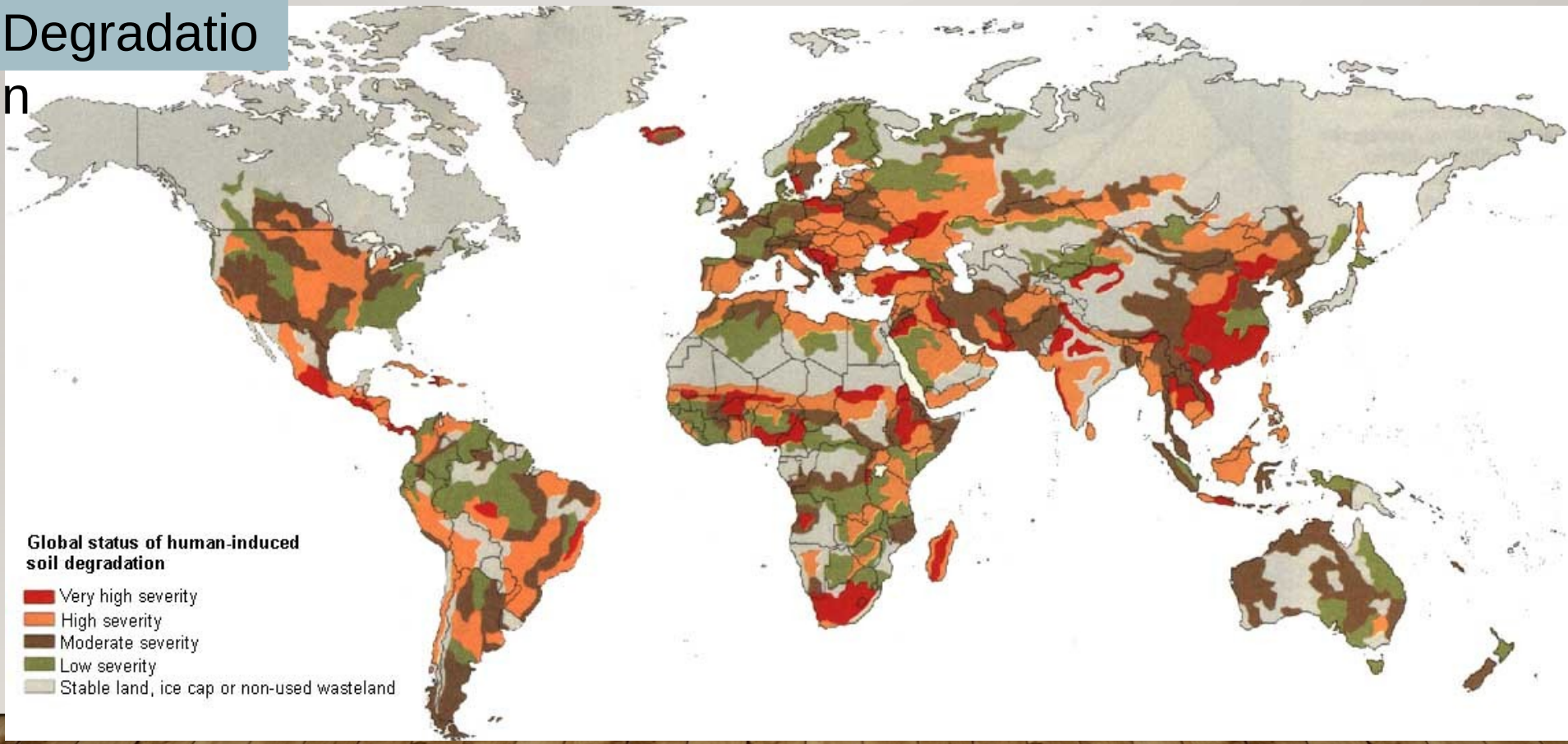
Impacts…

On farmlands, lower crop yields…. On ranchlands, fewer livestock can be supported per unit area…On nature reserves, lower biological diversity.

Global Environmental Concerns

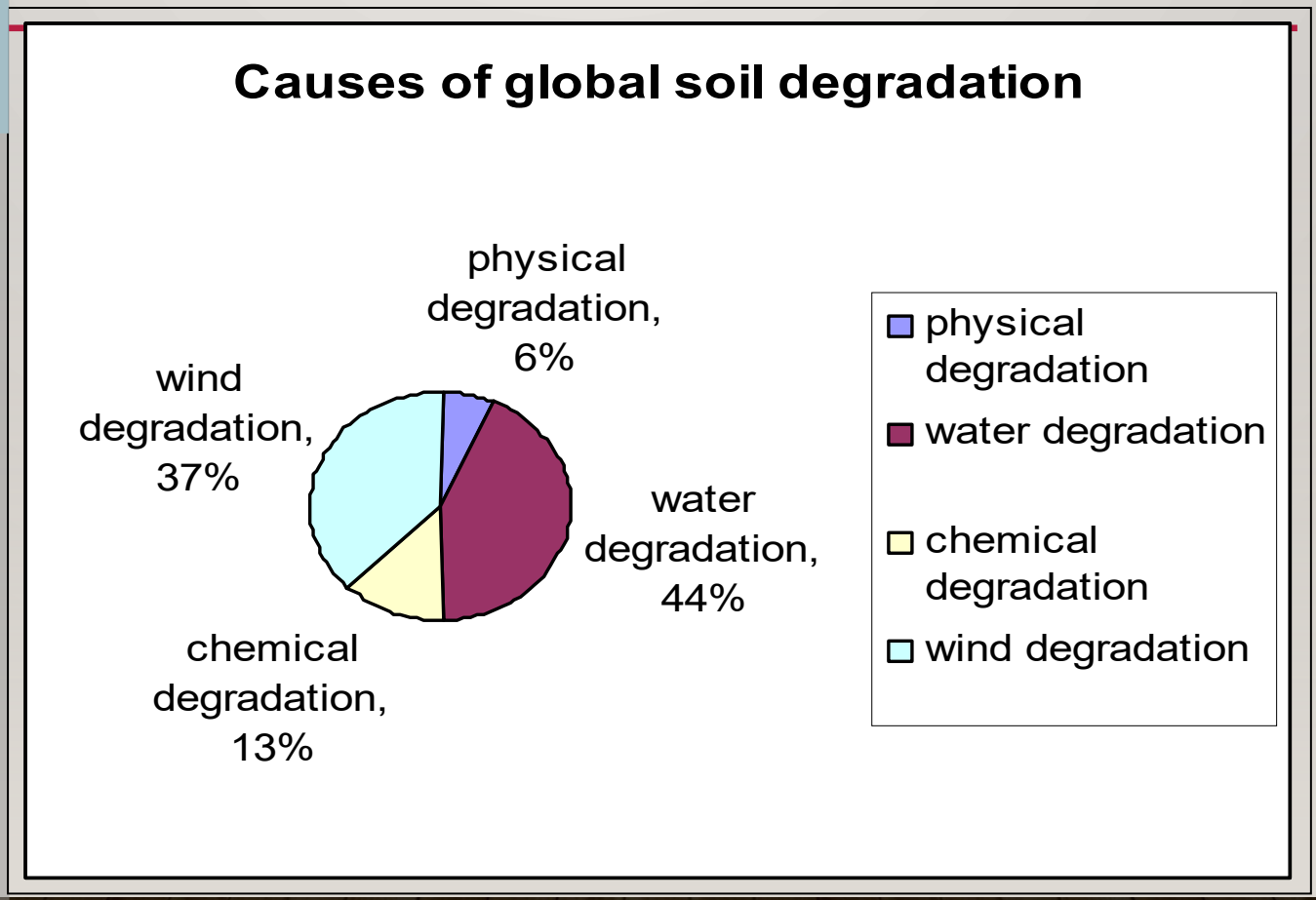
Human-induced soil degradation

Soil Degradation



Global Environmental Concerns

Soil Degradation



Soil Degradation Processes, Causes, and Assessment Approaches, Table 1 The global extent of human-induced soil degradation. (Modified from Oldeman et al. 1991)

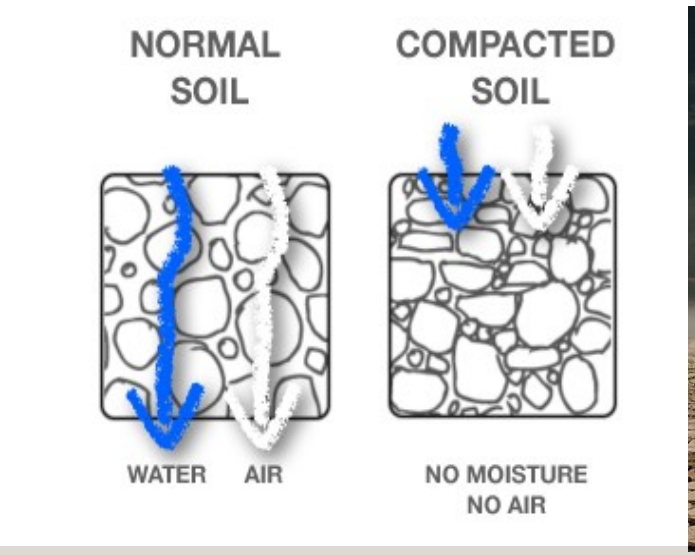
Region	Land surface (10^6 ha)		Land surface affected by different types of soil degradation (10^6 ha)			
	Total	Degraded	Water erosion	Wind erosion	Chemical soil degradation	Physical soil degradation
Africa	2,966	494	227	186	62	19
Asia	4,256	748	441	222	74	12
South America	1,768	243	123	42	70	8
Central America	306	63	46	5	7	5
North America	1,885	96	60	35	–	1
Europe	950	218	114	42	26	36
Oceania	882	102	83	16	1	2
World	13,013	1965	1,094	548	240	83

"–" indicates that an eligible area is affected by the specific degradation process

Global Environmental Concerns

Soil Degradation

Physical degradation includes compaction by heavy machinery or trampling by cattle, water accumulation from excess irrigation and poor drainage, and laterization (solidification of iron- and aluminum-rich tropical soil when exposed to sun and rain).





Global Environmental Concerns

Results…!

Soil Degradation



Water and wind erosion



Char dwellers in Bangladesh

Water induced erosion in Bangladesh

Global Environmental Concerns

Soil Degradation

Chemical degradation includes nutrient depletion, salt accumulation, acidification, and pollution



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Soil Degradation

sandy soil…less nutrient



SOIL DEGRADATION

- Deforestation: The main natural causes are fires and floods, and among human activities, the causes are logging, timber production, the conversion of a forest into agricultural land, and urbanization.
- Shifting cultivation: It is an old farming practice where the “slash-and-burn” technique is applied to clear land, followed by a long fallow period, which is important for the restoration of soil fertility
- Soil erosion is the main and most widespread form of land degradation. Soil erosion is caused by the activity of water and wind and represents the detachment and movement of soil particles from one place to another. This process can be natural or accelerated by human activity. It depends on many factors, among which the most important are the configuration of the terrain (slopes) and climatic-meteorological conditions.



Soil Degradation Processes, Causes, and Assessment Approaches, Fig. 1 Intensive erosion on the steep slopes of Stara Planina (*Old Mountain*), Eastern Serbia, caused by deforestation. (Author: Dragović 2007)



Soil Degradation Processes, Causes, and Assessment Approaches, Fig. 2 (**a**) A single concrete check dam constructed for sediment deposition, Salzburg, Austria, (**b**) a series of concrete check dams for slope and water energy reduction, Tyrol, Austria. (Author: Dragović 2008)

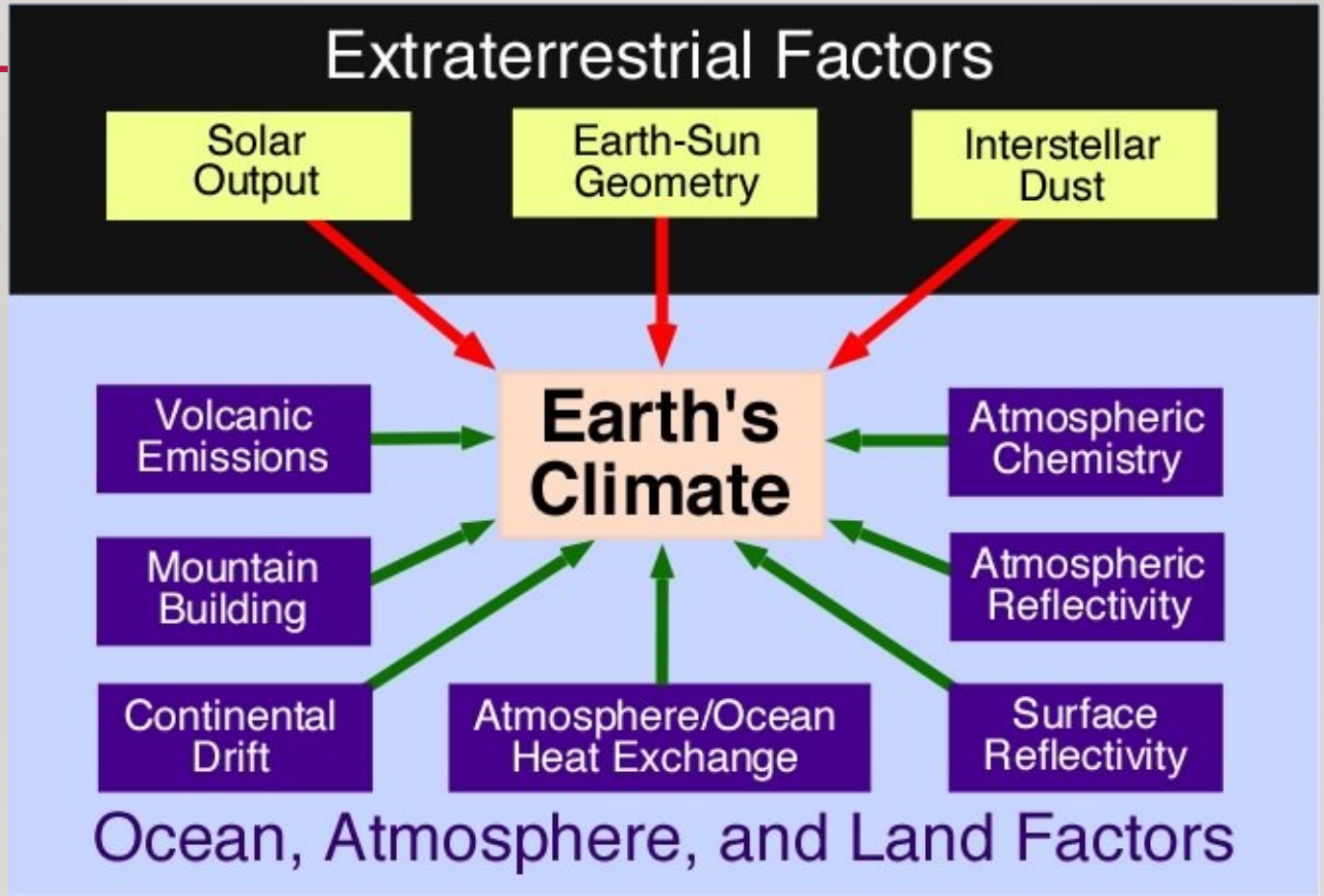
the structure, aeration, infiltration, water retention capacity, and soil biodiversity (Montanarella 2007). It serves as a soil acidity puffer and a source of energy for soil microorganisms as well. Organic carbon is the most important component (about 58%) of organic matter and soil quality indicators (Young et al. 2015). Significant carbon losses arise from activities that lead to the

- Salinization is one of the most widespread forms of soil degradation and occurs in arid and semi arid areas, where the amount of precipitation is small and irrigation is applied without a proper drainage system. However, salinization can occur in all climatic areas if irrigation is irregularly applied.
- Acidification, like salinization, is a severe form of degradation that results in a reduction in the agricultural land production potential. The acidification process prevents plants using water, resulting in drying and erosion, and causing increased leaching of nutrients and an irreversible breakdown of silicate minerals in the soil(Fitzpatrick 2002).

- Overgrazing, monocropping, and the use of agrochemicals, oil contamination from industry, mining, illegal landfills, and poorly managed landfills also contribute to the degradation of soil.
- **The assessment of soil degradation depends on the type of degradation process, the scale of the assessment, and a method that can be based on expert opinion (through questionnaires), remote sensing (satellite imagery), or modelling**

Global Environmental Concerns

Atmospheric Change



EXTRA-TERRESTRIAL FACTORS

- Solar output affects our climate—both directly and indirectly, **by changing the rate of solar heating of the Earth and atmosphere**, and indirectly, by changing cloud-forming processes.
- The Sun-Earth geometry **causes heating differences across the planet**. Different parts of the Earth experience variations in climatic changes due to the change in the Earth's position caused by revolution around the Sun. The four seasons are summer, winter, fall, and spring.

- **Interstellar dust** in the atmosphere primarily affects Earth's climate by scattering and absorbing **shortwave and longwave radiation,** directly influencing surface temperature. Beyond that, it affects life on Earth through direct and indirect interactions with the biotic and abiotic components of the Earth system. Dust affects the very basic process of rainfall through its direct and indirect effects; thus, it regulates the hydrological cycle and the monsoon systems, which ultimately affect the distribution of water resources and agricultural production

TERRESTRIAL FACTORS

- Atmospheric chemistry plays a critical role in climate change by **controlling the amount and distribution of important climate forcing agents like water vapor, methane, and tropospheric and stratospheric ozone**.
- Atmospheric reflectivity: When sunlight reaches Earth, it can be reflected or absorbed. The amount that is reflected or absorbed depends on Earth's surface and atmosphere. Any change in surface albedo will alter climate by **drastically changing the amount of solar energy absorbed by the planet**. When the angle of the sun to the surface is low (closer to the horizon), solar energy is less intense since it is spread out over a larger area. The term **albedo** refers to the amount of solar radiation reflected from an object or surface, often expressed as a percentage. Earth has an albedo of about 30%, meaning that 70% of the sunlight that reaches the planet is absorbed. Absorbed sunlight warms Earth's land, water, and atmosphere.

- Surface Reflectivity: **The capability of a surface to reflect solar energy** is measured by its albedo or reflectivity in the visible portion of the energy spectrum. **Light-colored objects and surfaces, like snow and clouds, tend to reflect most sunlight,** while **darker objects and surfaces, like the ocean, forests, or soil, tend to absorb more sunlight.**

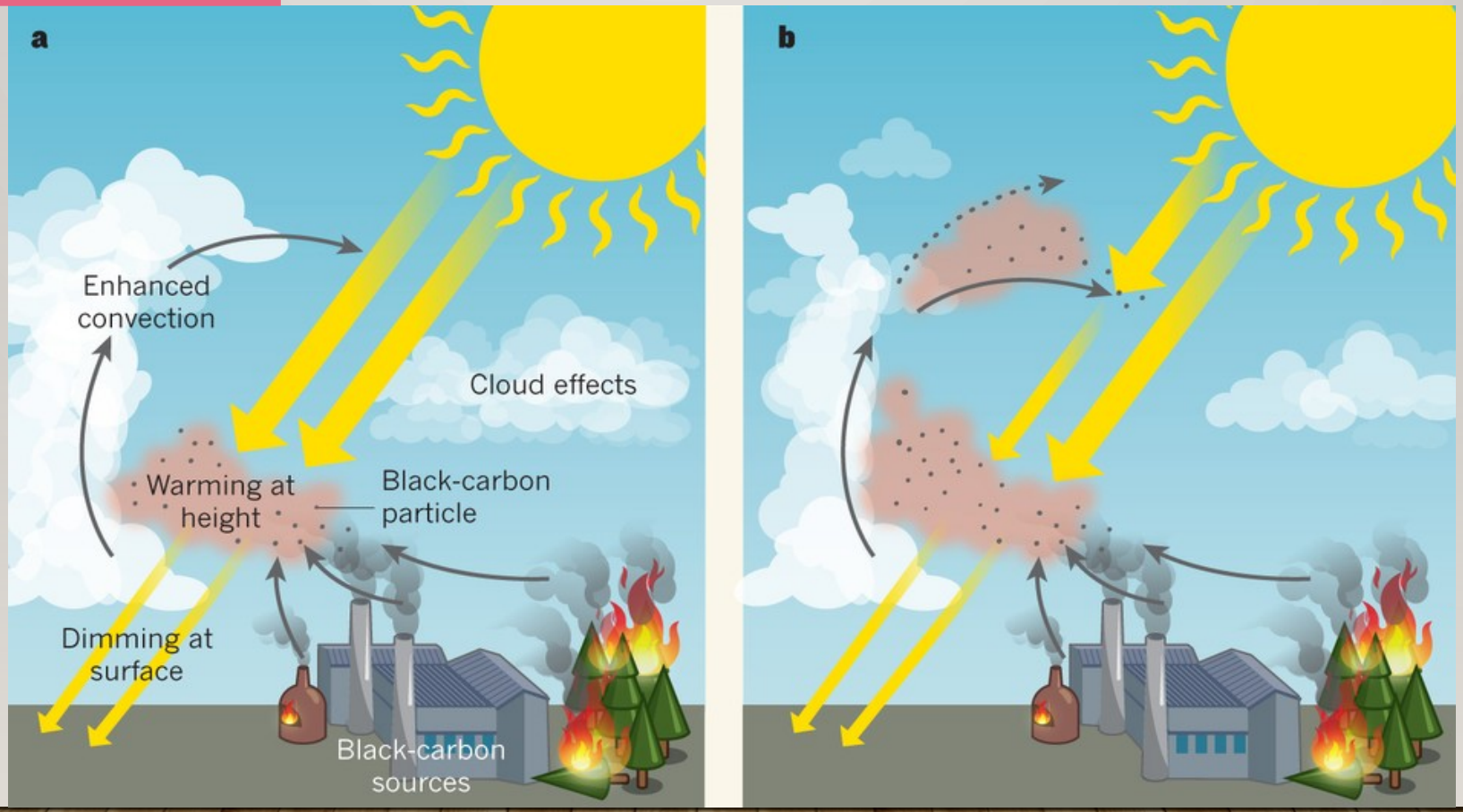
- Continental drift is the hypothesis that the Earth's continents have moved over geologic time relative to each other, thus appearing to have "drifted" across the ocean bed.
- The idea of continental drift has been subsumed into the science of plate tectonics, which studies the movement of the continents as they ride on plates of the Earth's lithosphere

- Atmosphere/ocean heat exchange: the oceans impact weather and climate by heating (and cooling) the lower atmosphere. As **seawater evaporates, the ocean surface cools; when the moisture later condenses into cloud droplets, this heat is released, warming the atmosphere**.
- Mountain building refers to the **geological processes that underlie the formation of mountains.** These processes are associated with large-scale movements of the Earth's crust (tectonic plates). Folding, faulting, volcanic activity, igneous intrusion, and metamorphism can all be parts of the orogenic process. The formation of mountains is not necessarily related to the geological structures found in them.
- Volcanic gases like sulfur dioxide can cause global cooling, while volcanic carbon dioxide, a greenhouse gas, **has the potential to promote global warming.**

Global Environmental Concerns

Atmospheric Change

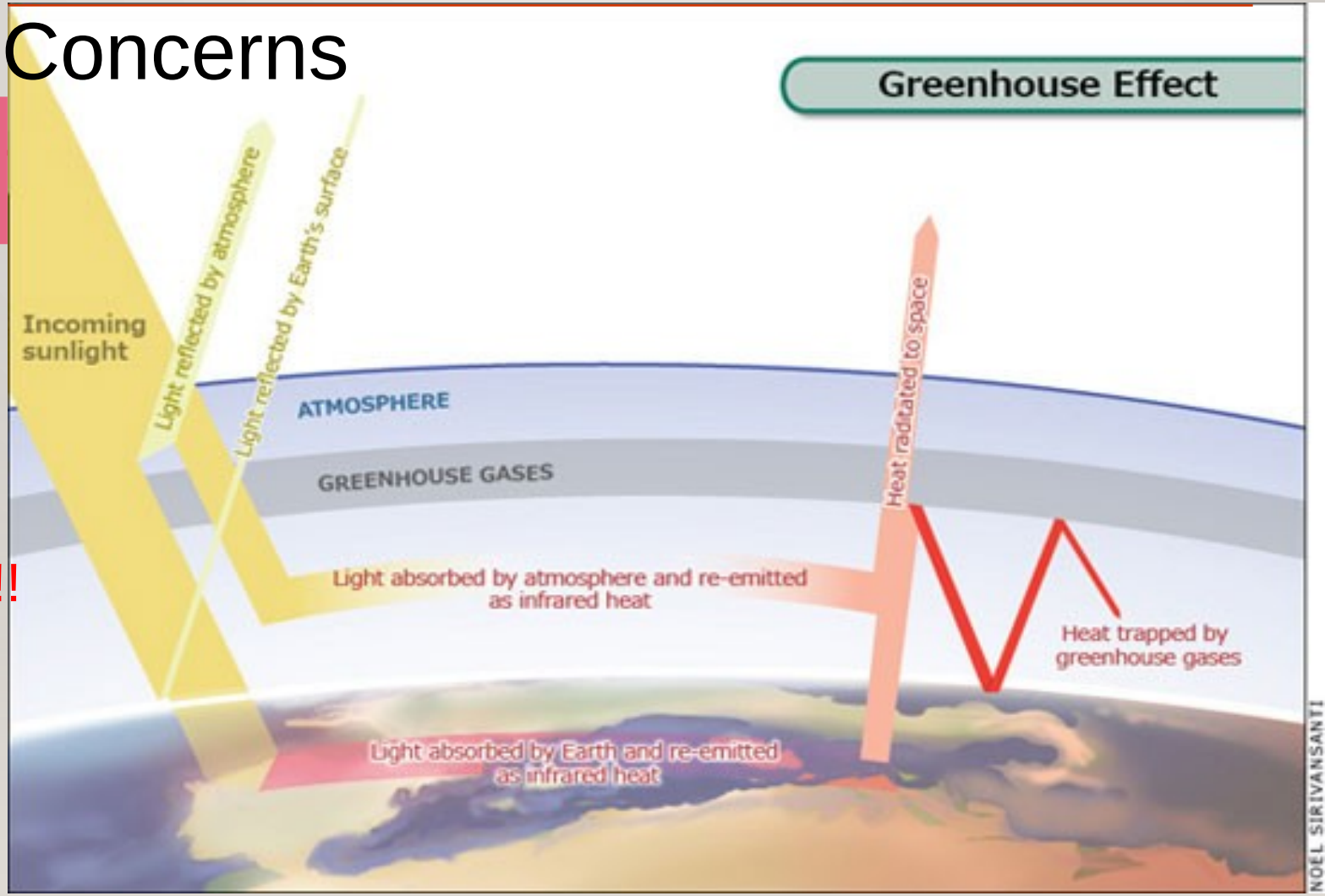
Human-Induced Factors



Global Environmental Concerns

Atmospheric Change

Our Concern!!!



Global Environmental Concerns

Impacts…!!!

Atmospheric Change

Increased global temperature

At the current rate, global temperature may increase by 1.8–4.0ºC by the end of this century (IPCC 2007)

Greenhouse gases can influence animal populations through mechanisms

… can enter terrestrial and aquatic food webs and alter ecosystem functioning

Atmospheric contaminants commonly do this in two ways: 1) bottom-up and 2) top-down pathways

1

Alter plant anatomy and physiology

2

Alter population dynamics for their prey

Global Environmental Problems

Biodiversity loss

Biodiversity loss

Defined as…

«*The long-term or permanent quantitative or qualitative reduction in components of biodiversity and their potential to provide goods and services, to be measured at global, regional and national levels».*

(Convention on Biological Diversity)

Loss of diversity (e.g., species, genetic, ecosystem, etc.),

Loss/diminished capacity for the components of diversity to provide a particular service (e.g., unsustainable harvest),

Homogenization of biodiversity (e.g., monoculture agriculture).

WHY IS BIODIVERSITY LOSS A CONCERN?

LIFE HAS EXISTED ON EARTH FOR OVER 3.5 BILLION YEARS. OVER 95% OF THE SPECIES THAT EVER EXISTED HAVE GONE EXTINCT. WHY



Why is biodiversity loss a concern?

Human Well-being
Basic material for a good life
Health
Security
Good social relations
Freedom of choice and action

Ecosystem Services

Provisioning Services
Food, fiber, and fuel
Genetic resources
Biochemical
Fresh Water

Cultural Services
Spiritual and religious values
Knowledge system
Education and inspiration
Recreation and aesthetic values
Sense of place

Supporting Services
Primary production
Provision of habitat
Nutrient Cycling
Soil formation and retention
Production of atmospheric oxygen
Water cycle

Regulating Services
Invasion resistance
Herbivore
Pollination
Seed dispersal
Climate regulation
Disease regulation
Natural hazard protection
Erosion regulation
Water purification
Pest Regulation

Biodiversity
Number
Relative abundance
Composition
Interactions

Ecosystem Functions

Millennium ecosystem assessment, 2005

Example: Provisioning services



Example: Provisioning services



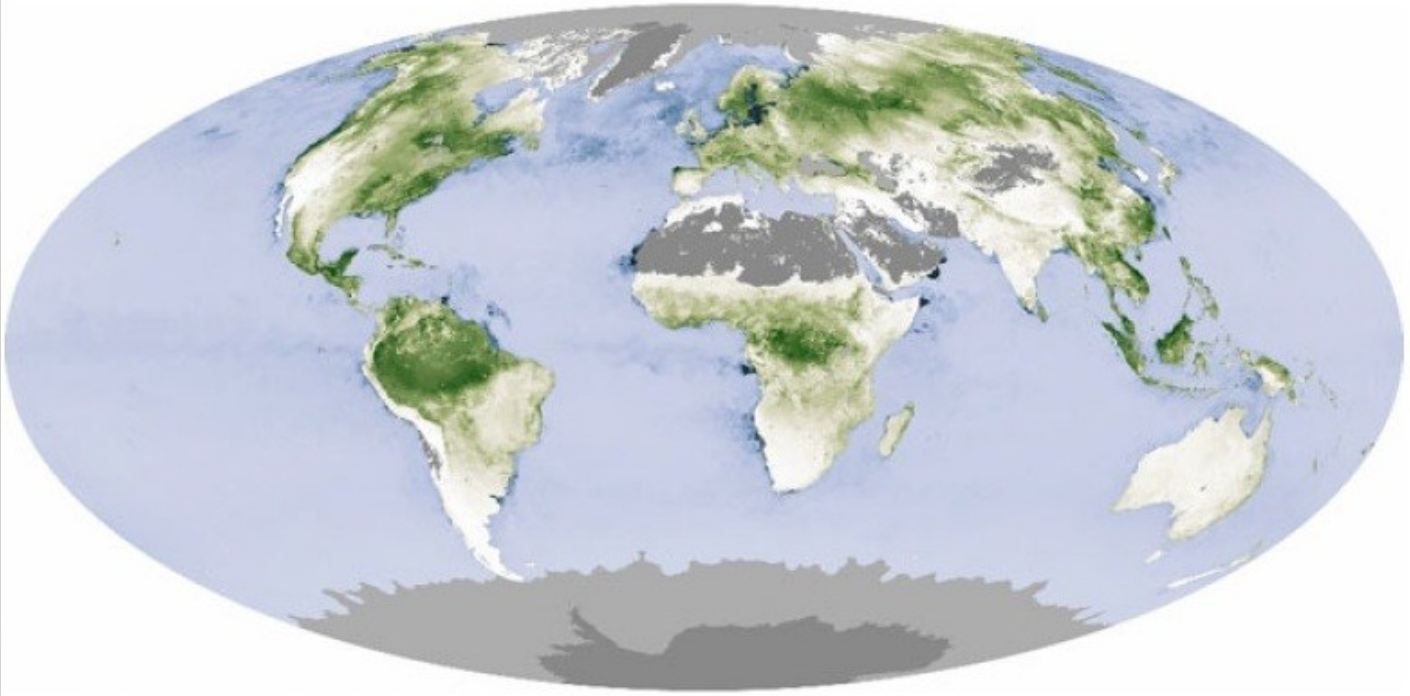
Example: Cultural services



Example: Cultural services



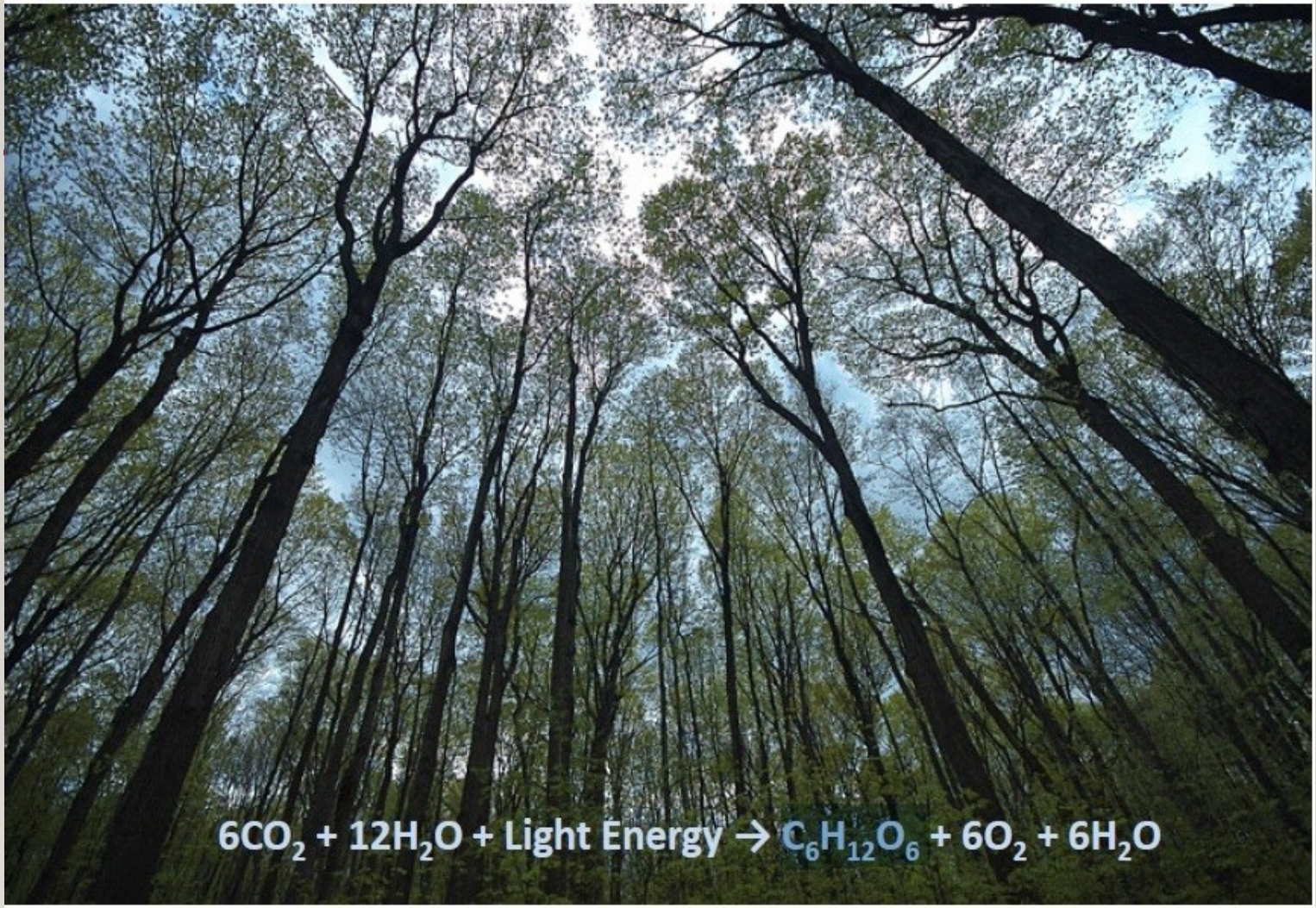
Example: Supporting service



Land and ocean net primary production estimated from spectral data

Source: NASA Earth Observatory

Example: Supporting service



Example: Regulating services







Example: Regulating service



Thanks

References

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